



Multimedia Security and Privacy

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Outline

Usages of Digital Watermarking

Watermarking vs Data Hiding

Watermark detection

Watermark detection in practice

Multiple bit data hiding

Multi-bit spread spectrum watermarking

Perceptual masking

Embedding domain

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Watermarking

Definition

Ensures ownership and authenticity of digital content

Requirements

- Robustness
- Percebility
- Embedding Capacity ($\sim 64 - 128$ bits)

Applications

- Copyright protection
- Real vs synthetic media detection

Steganography

Definition

Hides the existence of a message for secret communication

Requirements

- Undetectability
- Embedding Capacity (~ 1 Ko per image)

Applications

- Alternative to cryptography



Tamper-proofing

Definition

Detects unauthorized modifications to digital content (Key based or compression-based)

Requirements

- Capacity
- Robustness
- Localization

Applications

- Integrity verification
- Forensics

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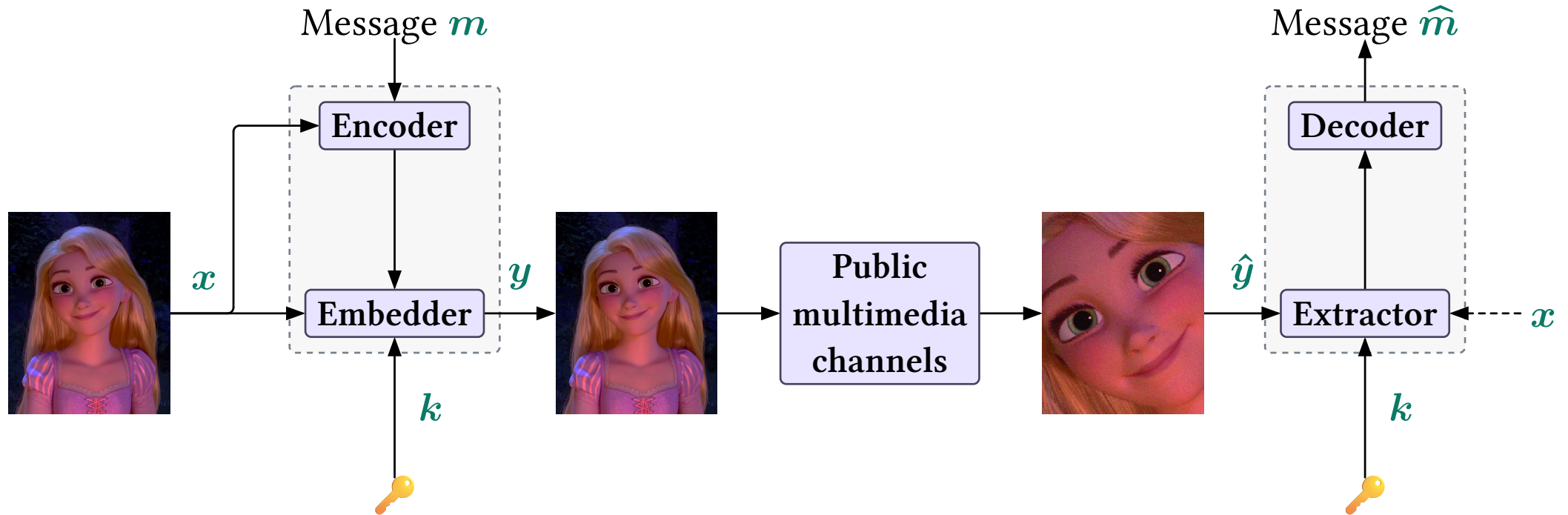
Watermarking vs Data Hiding

Technologies

- Zero-bit watermarking (Detection problem)
- Multi-bit watermarking (Decoding problem)



Block diagram





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Watermark detection

Hypothesis testing

- $H_0 : y = x$
- $H_1 : y = x + w$

Score definition

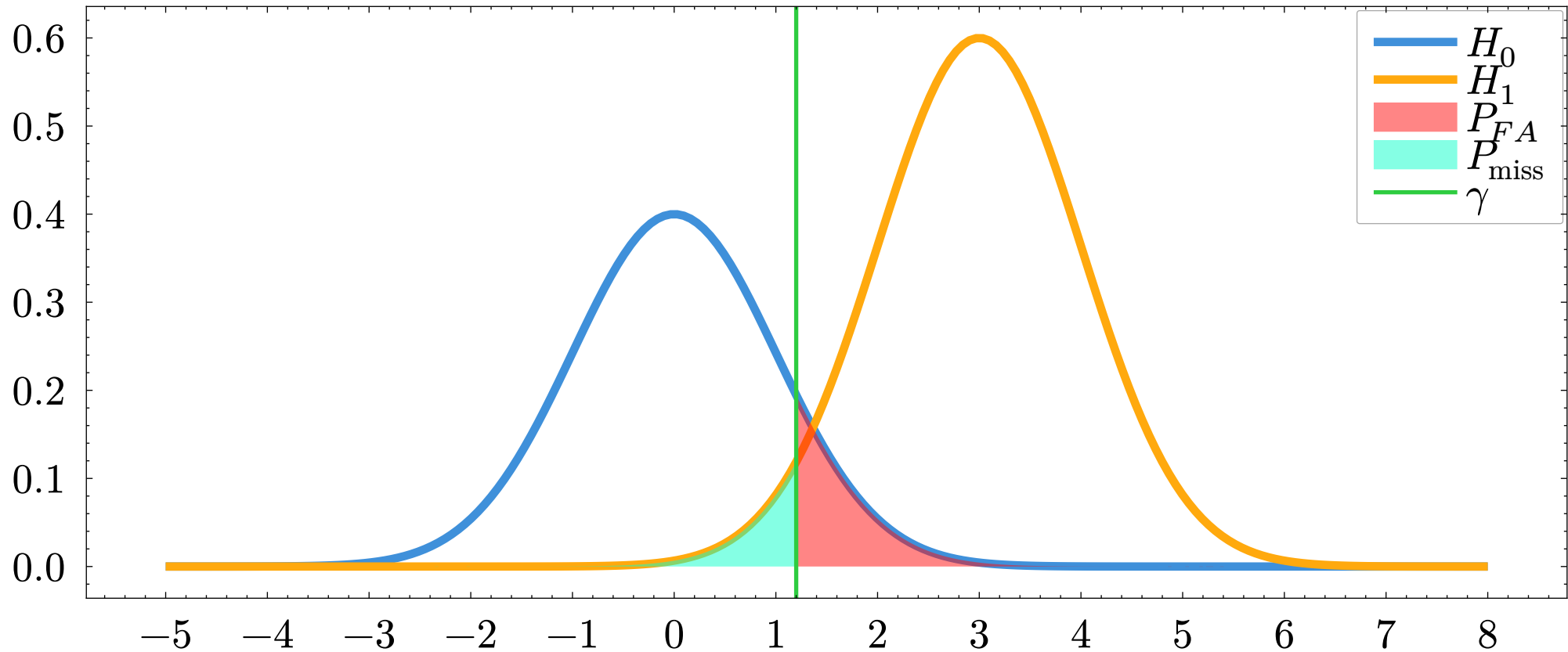
- $S = \frac{y^T w}{M}$ (M : number of pixels)
- $S \sim \mathcal{N}(\mu_{H_0}, \sigma_{H_0}^2)$ under H_0 (Central Limit Theorem)
- $S \sim \mathcal{N}(\mu_{H_1}, \sigma_{H_1}^2)$ under H_1

Probabilities

- Correct Detection: $P_{D(\gamma)} = \Pr(S \geq \gamma \mid H_1)$
- False Acceptance: $P_{FA}(\gamma) = \Pr(S \geq \gamma \mid H_0)$



Graph hypothesis testing



Graph hypothesis testing

Neyman Pearson Strategy

- Fix P_{FA} :

$$P_{FA}(\gamma) = \int_{>\gamma} p(x \mid H_0) dx$$

- Minimize probability of miss:

$$P_{\text{miss}} = 1 - P_D = \int_{\leq \gamma} p(x \mid H_1) dx$$

Bayesian Strategy

- Fix priors $P(H_0)$ and $P(H_1)$
- Minimize overall error: $P_{\text{error}} = P_{FA}P(H_0) + P_{\text{miss}}P(H_1)$

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Cosine similarity

- $\cos(\theta) = \frac{y^T w}{\|y\|_2 \|w\|_2}$

Score definition

- $S(y) = \frac{1}{M} (y^T w) = \frac{1}{M} \|y\|_2 \|w\|_2 \cos(\theta)$

Distribution under H_0

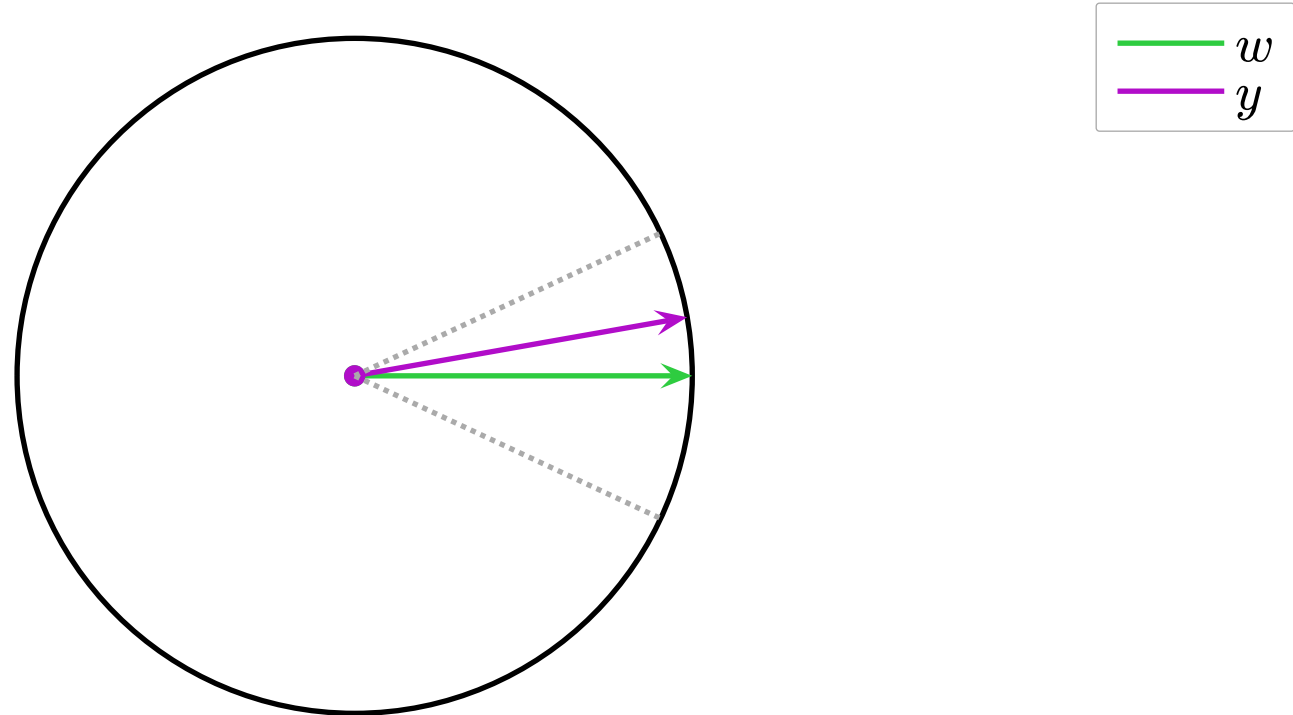
- $S(y) = \frac{1}{M} x^T w \sim \mathcal{N}\left(0, \frac{\|w\|_2^2 \sigma_x^2}{M^2}\right)$

Distribution under H_1

- $S(y) = \frac{1}{M} (x + w)^T w \sim \mathcal{N}\left(\frac{\|w\|_2^2}{M}, \frac{\|w\|_2^2 \sigma_x^2}{M^2}\right)$

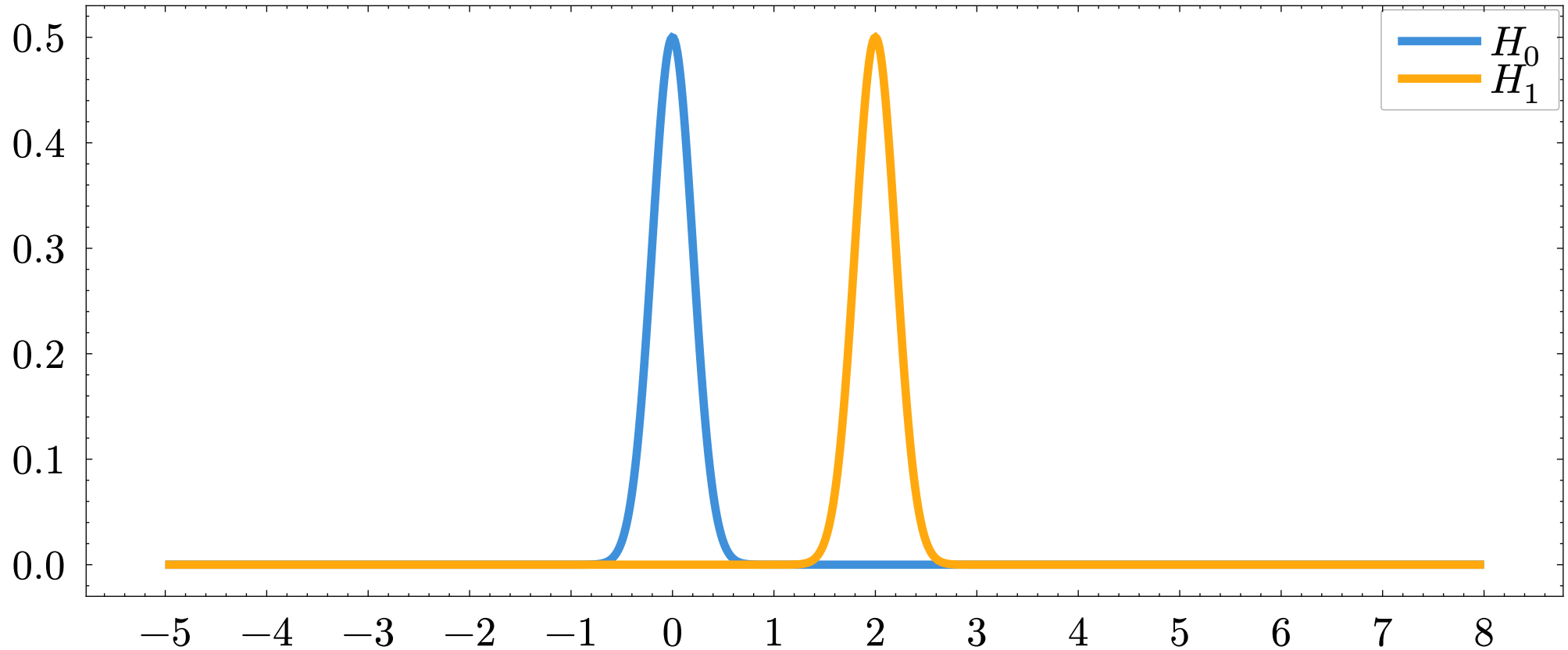


Decision rule: $|y^T w| > \|y\|_2 \|w\|_2 \cos(\theta)$



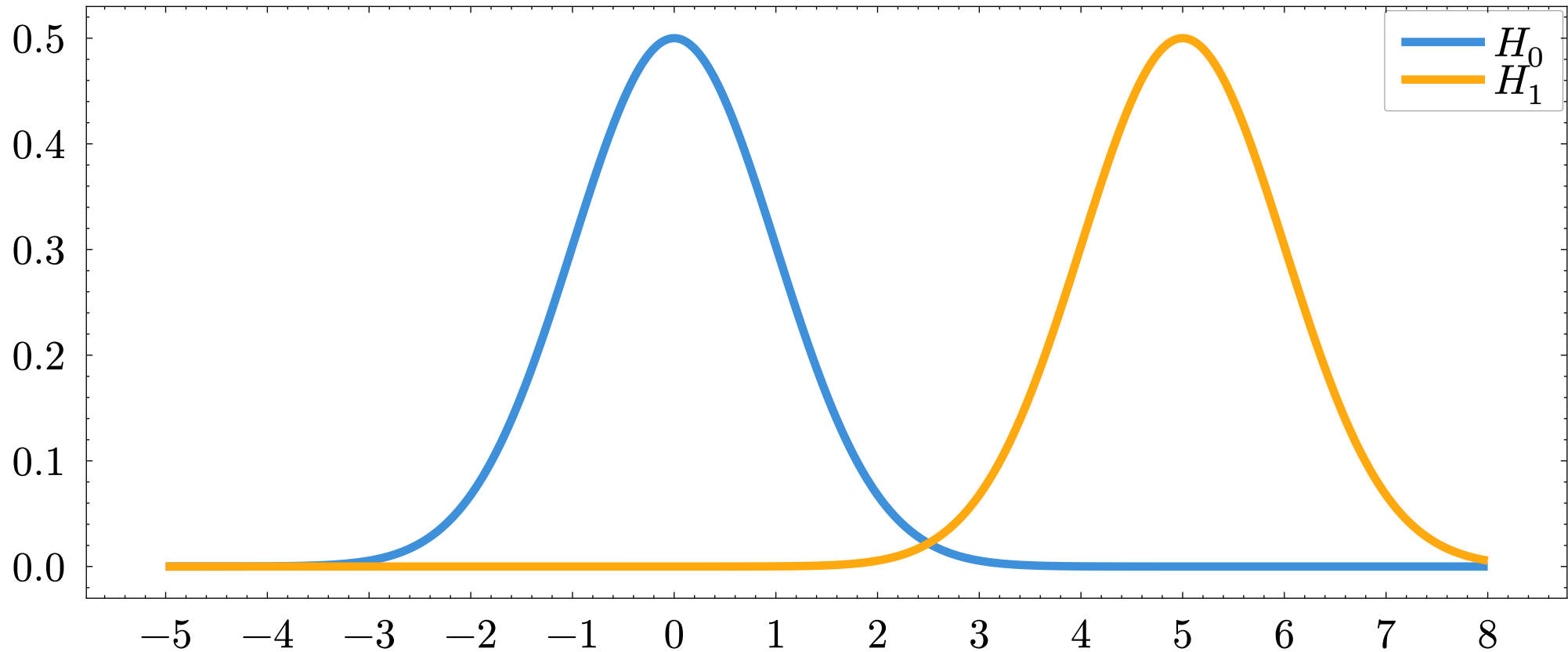


M increases



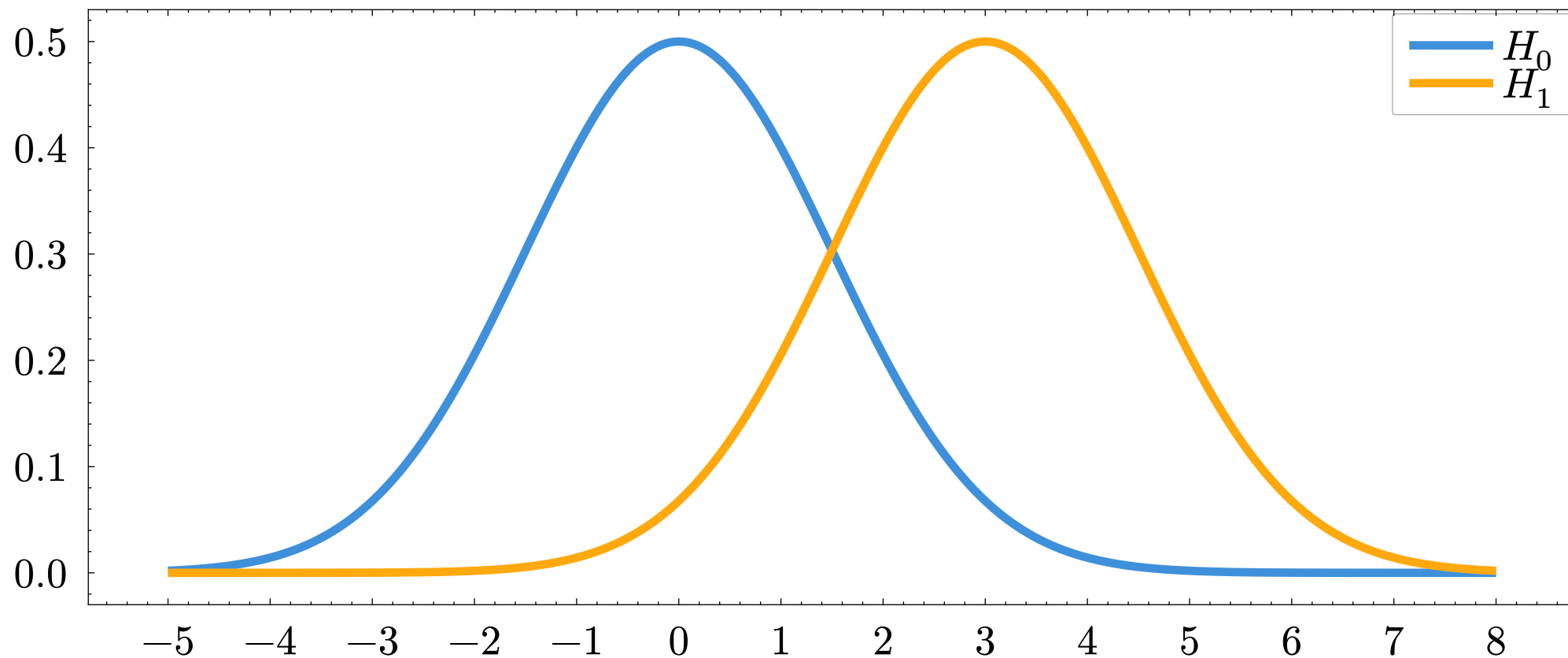


$\|w\|_2^2$ increases





σ_x^2 increases



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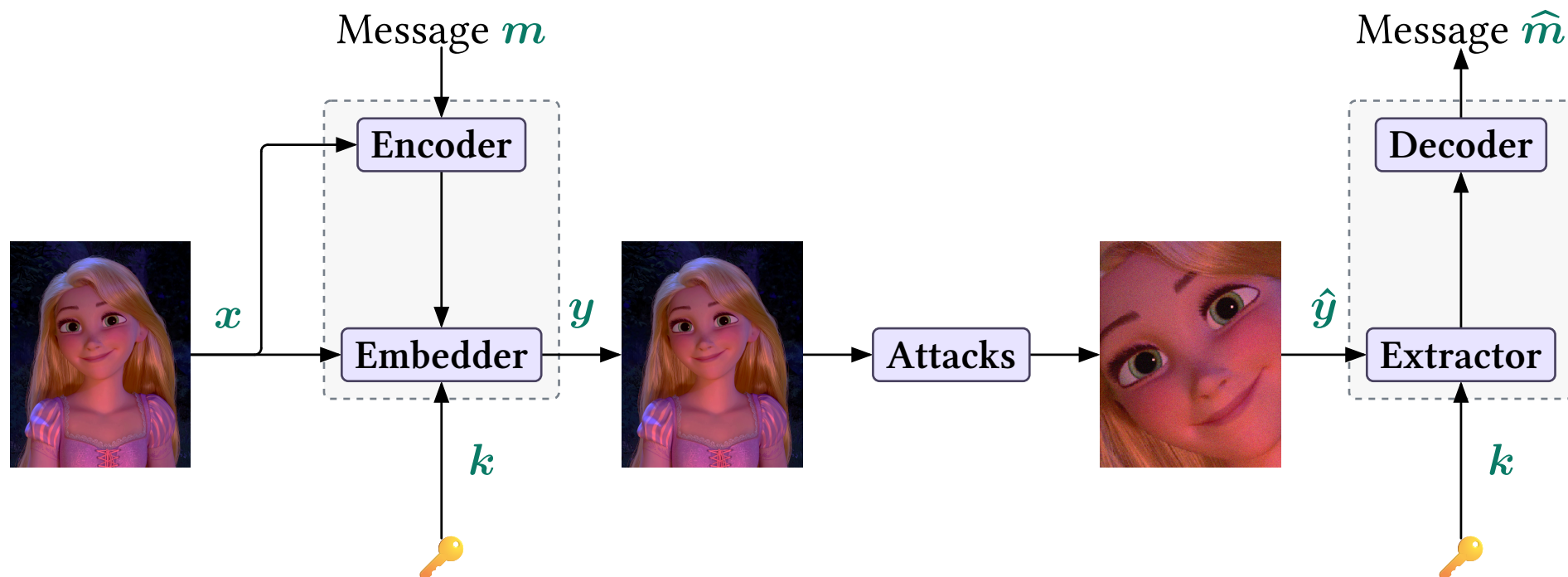
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Multiple bit data hiding



Random vs Periodic

Random

- Pseudo-random embedding positions determined by a key
- More secure against targeted removal
- More vulnerable to geometric attacks

Periodic

- Embedding at fixed intervals (e.g., every 8 pixels)
- More robust to geometric attacks (self-synchronization)
- More vulnerable to targeted removal

Error correction (Repetition-Based and ECC encoding)

- Increase robustness



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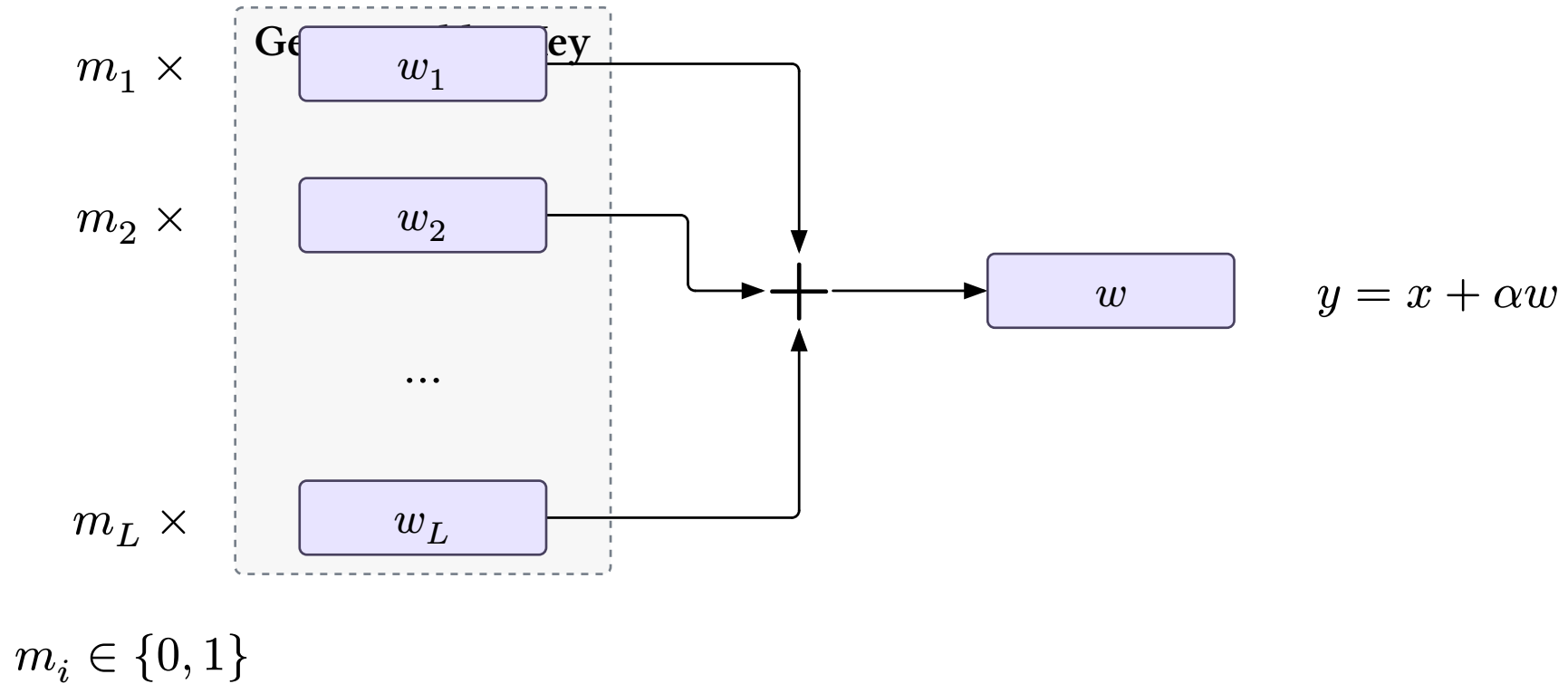
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Main requirements for privacy protection



Multi-bit spread spectrum watermarking

Main requirements for privacy protection



Detection

Main requirements for privacy protection

- Each carrier is orthogonal to the others:
 - $w_i^T w_j = 0$ for $i \neq j$
 - $w_i^T w_i = \|w_i\|_2^2 > 0$
- Each carrier is detected independently:
 - $S_i = y^T w_i$
 - Taking the sign of S_i gives the bit value $\widehat{m}_i$
- Link to zero-bit detection:
 - Similar to performing L independent zero-bit detections, one for each carrier



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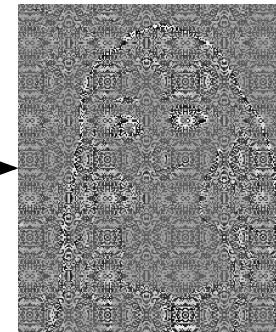
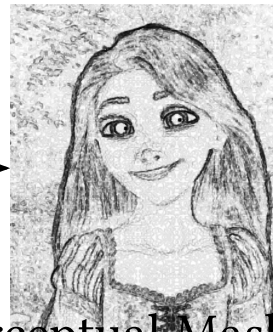
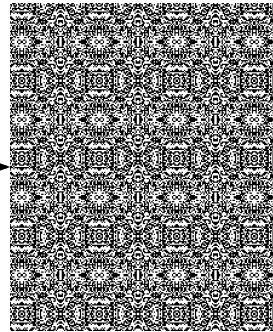
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Perceptual masking

No Perceptual Masking



Perceptual Masking



Perceptual masking

Local Variance

$$\sigma^2(x) = \frac{1}{W^2} \sum_{i=1}^W \sum_{j=1}^W x_{ij}^2 - \left(\frac{1}{W^2} \sum_{i=1}^W \sum_{j=1}^W x_{ij} \right)^2$$

Noise visibility function

$$\text{NVF} = \frac{1}{1 + D \cdot \frac{\sigma^2(x)}{\sigma_{\max}^2}}$$



Perceptual masking

Weighted Mean Squared Error

$$\text{WMSE} = \frac{1}{H \cdot W} \sum_{i=1}^H \sum_{j=1}^W \text{NVF}_{ij} (x_{ij} - y_{ij})^2$$

Weighted Peak Signal to Noise Ratio

$$\text{WPSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{WMSE}} \right)$$

Embedding with perceptual masking

$$y = x + \alpha_1 \text{NVF} \cdot w + \alpha_2 (1 - \text{NVF}) \cdot w \quad \text{with } \alpha_1 < \alpha_2$$



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Embedding domain

Spatial domain

- Directly modify pixel values
- Fast and simple

Frequency domain

- Modify coefficients in a transform domain (e.g., DCT, DFT, DWT)
- Sensitive to affine transformations
- Good for compression robustness
- Fast but requires transform computation

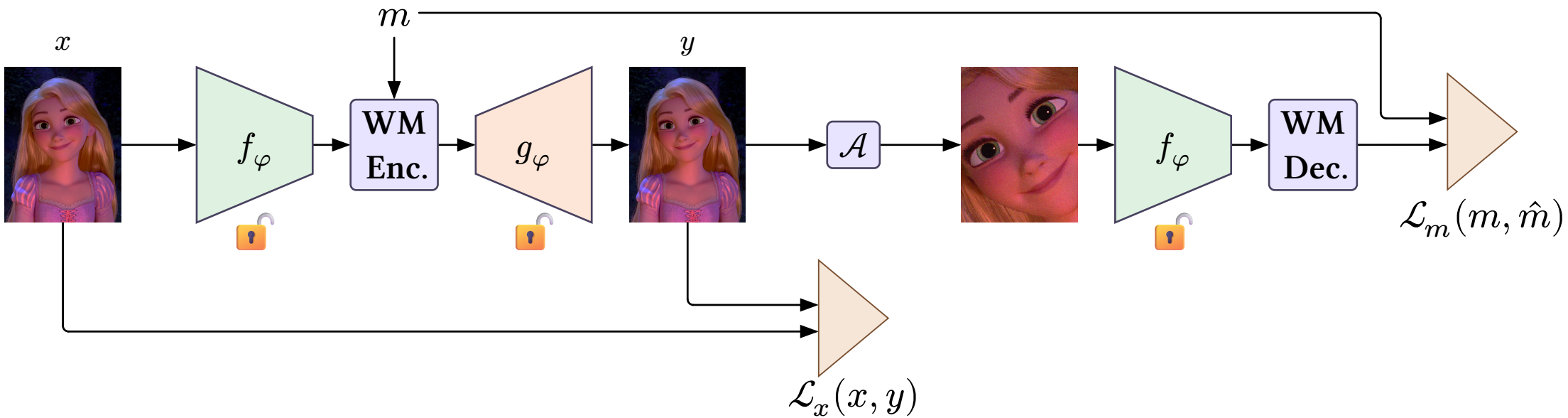
Embedding domain

Learnable domain

- Linear (e.g., PCA)
 - Energy compaction
 - Linear => limited robustness to complex transformations
- Non-linear (e.g., Autoencoders)
 - Robust to noise and minor geometric distortions
 - Flexible and adaptable to different requirements
 - But requires training and tuning
- Foundation models (e.g., CLIP)
 - High robustness to a wide range of transformations
 - Allows to exploit pretrained models without extensive retraining
 - Computationally expensive and vulnerable to adversarial attacks

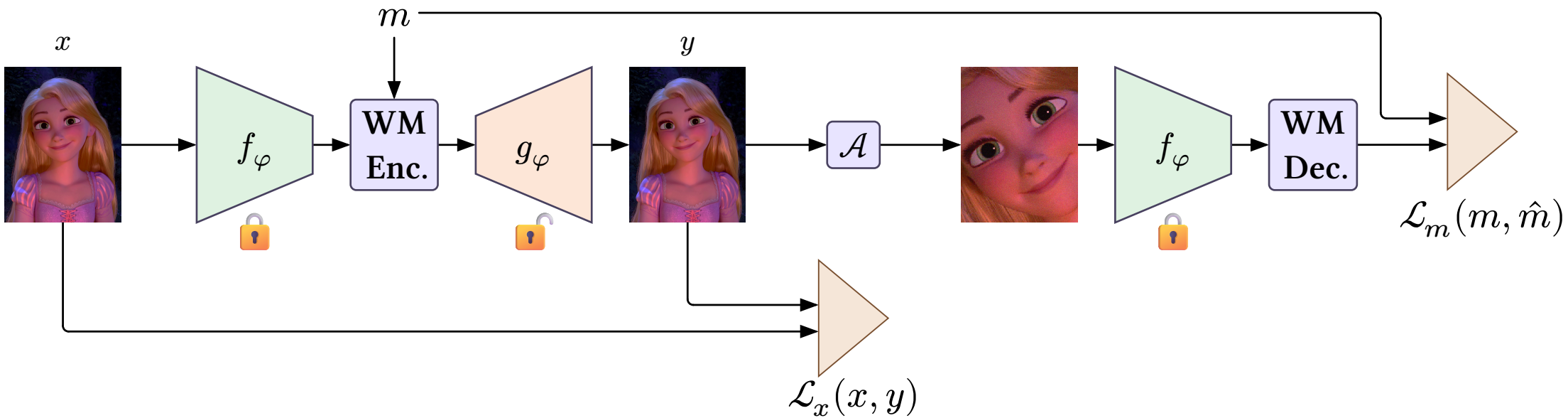


Autoencoder-based watermarking





Foundation model-based watermarking



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Geometrical synchronization

Hand-crafted invariant domain watermarking

- Embed in a domain invariant to geometric transformations
 - Fourier (translation), Fourier-Mellin (rotation, scale, translation)

Feature-based synchronization

- Use robust features (e.g., SIFT) (requires exact prior knowledge)

Self-synchronizing watermarking

- Repeated structure (security leak, size constraints, perceptual visibility)

Foundation model-based synchronization

- Use pretrained invariant models (e.g., CLIP, DINO) (adversarial + computational cost)



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Additive vs Quantization approaches

Additive

- Simple and flexible
- Sensitive to host interference

Binary Quantization Index Modulation

- More robust to noise
- Sensitive to scaling and volumetric operations

Additive vs Quantization approaches

Vector Quantization

- Extension of QIM to multiple bits
- Better performance but more complex

Product Quantization

- Balances robustness and complexity
- Suitable for large-scale systems



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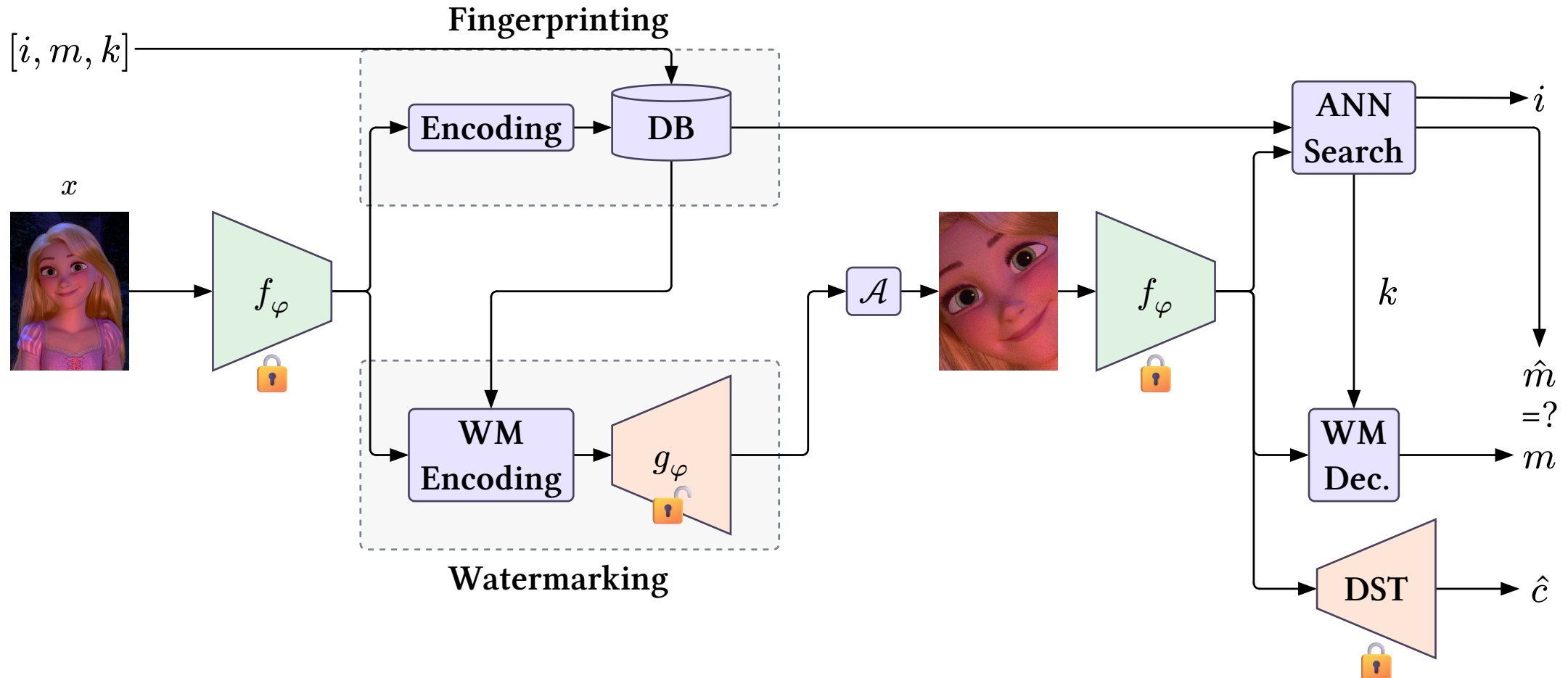
Main requirements for privacy protection

Key management

- Symmetric: Same key for embedding and extraction
- Key reuse: adversary can learn/estimate the key
- Kind of key:
 - positional key: determines embedding positions
 - encoding key: determines the codebook or carriers
 - XOR key: added to the message before encoding
- Problem: which key to use ?
 - Fingerprinting: category of input determines the key
 - Key-message search: find the key that decodes a meaningful message



Key management





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Transform domain watermarking

Discrete Cosine Transform (DCT)

- Commonly used in JPEG compression
- Good for robustness against compression
- Sensitive to geometric transformations

Discrete Wavelet Transform (DWT)

- Multi-resolution analysis
- Good for robustness against noise and minor geometric distortions
- Computationally more complex than DCT



Transform domain watermarking

Discrete Fourier Transform (DFT)

- $F(u, v) = \sum_{x=1}^M \sum_{y=1}^N f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$
- Invariant to translation
- Sensitive to rotation and scaling
- Computationally expensive

Fourier-Mellin Transform

- Log-polar mapping of the Fourier transform (degrades quality)
- $(u, v) \rightarrow \left(\log \sqrt{u^2 + v^2}, \tan^{-1} \left(\frac{v}{u} \right) \right) + 1\text{D Fourier in angular direction}$
- Invariant to rotation, scaling, and translation
- Good for robustness against geometric transformations
- Computationally expensive and less common in practice

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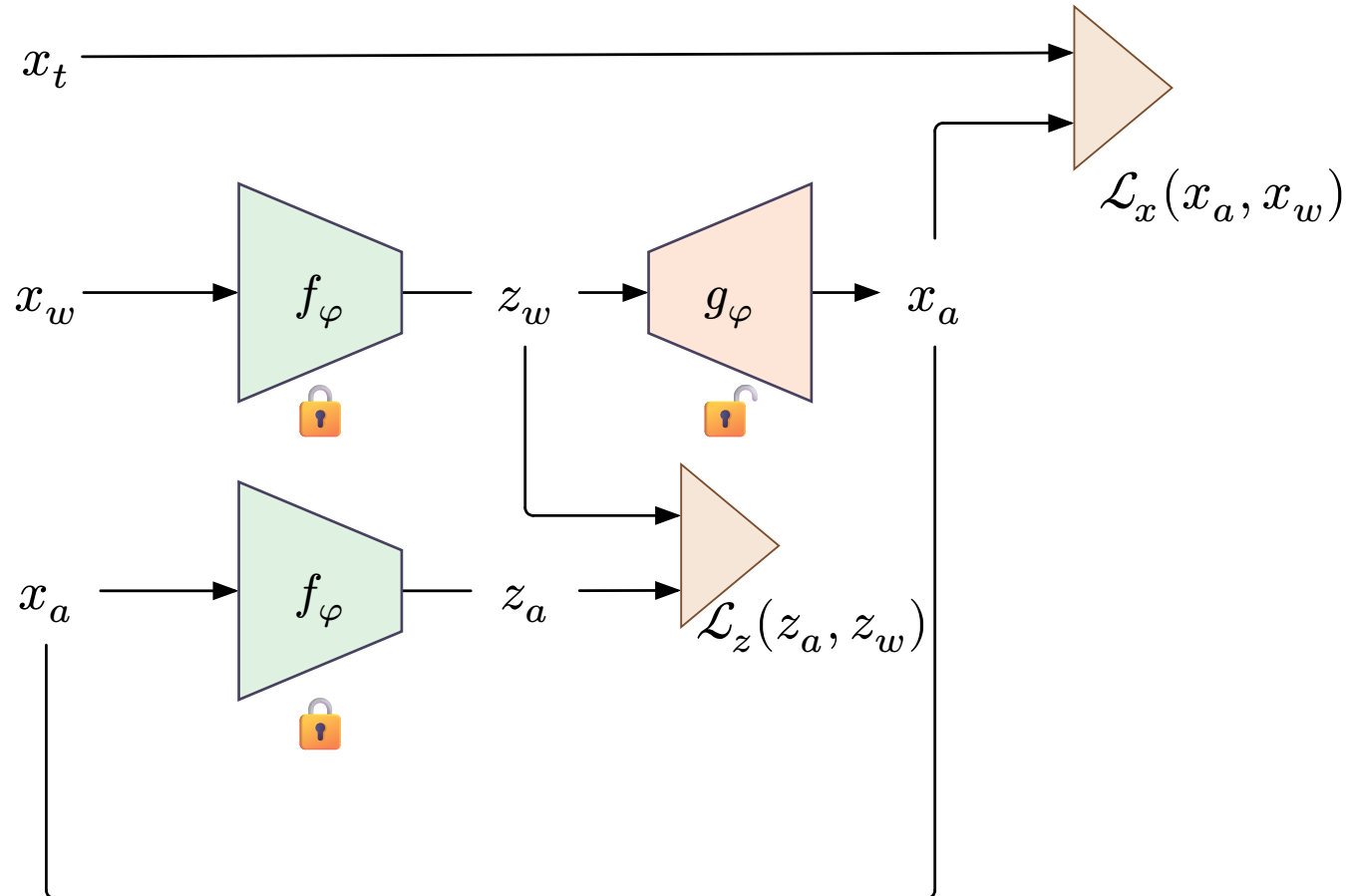
Removal attacks

- Denoising and noise addition
- Averaging
- Collusion (estimate watermark from each image -> align -> suppress)

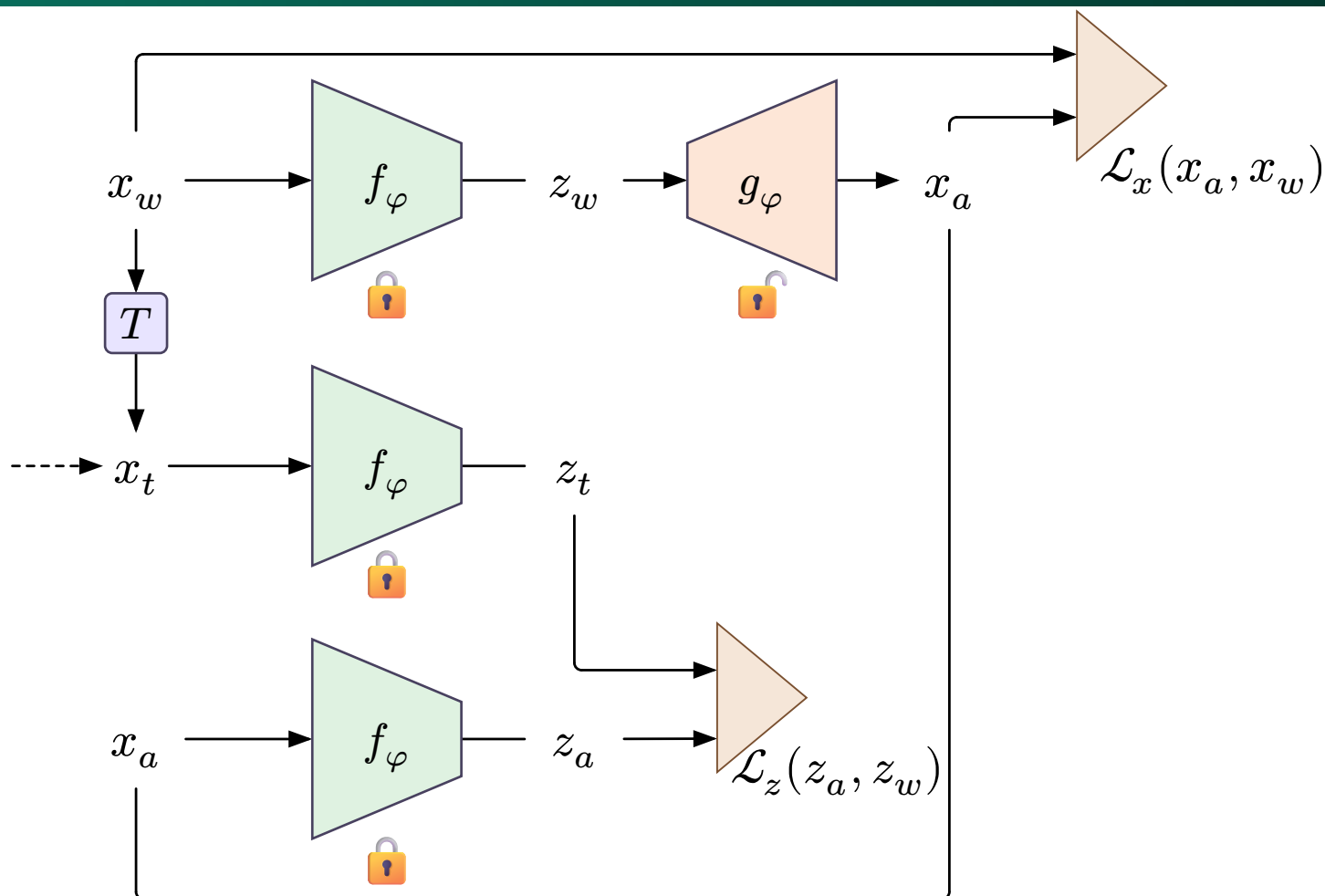
Copy attacks

- Copy the watermark from one to another image

Adversarial copy attack



Adversarial removal attack



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Cryptographic hashing

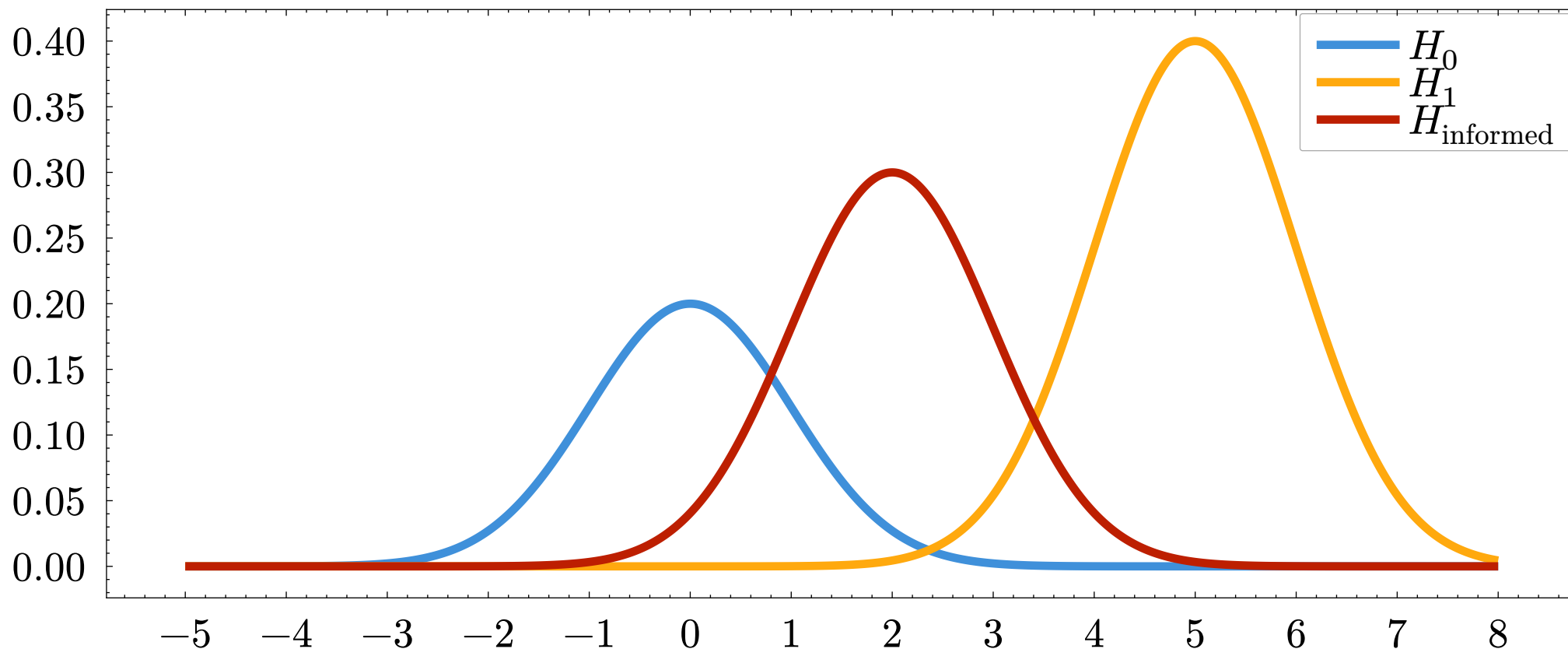
- Designed to be collision-resistant and unpredictable
- Small changes in input lead to large changes in output (avalanche effect)
- Not suitable for perceptual similarity tasks

Perceptual hashing

- Designed to produce similar outputs for perceptually similar inputs
- Robust to minor modifications (e.g., resizing, compression)
- Used for tasks like image retrieval and copyright enforcement
- Hypothesis testing:
 - $H_0: y = x + \varepsilon$
 - $H_1: y = x(m) + \varepsilon$



Perceptual vs cryptographic hashing



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Content fingerprinting

Definition

- Extract a unique identifier (fingerprint) from the content
- Used for tracking and identifying content without modifying it
- Challenges:
 - Robustness to transformations and attacks
 - Uniqueness and collision resistance
 - Privacy concerns

Applications

- Copyright
- Image and video identification
- Content-based retrieval

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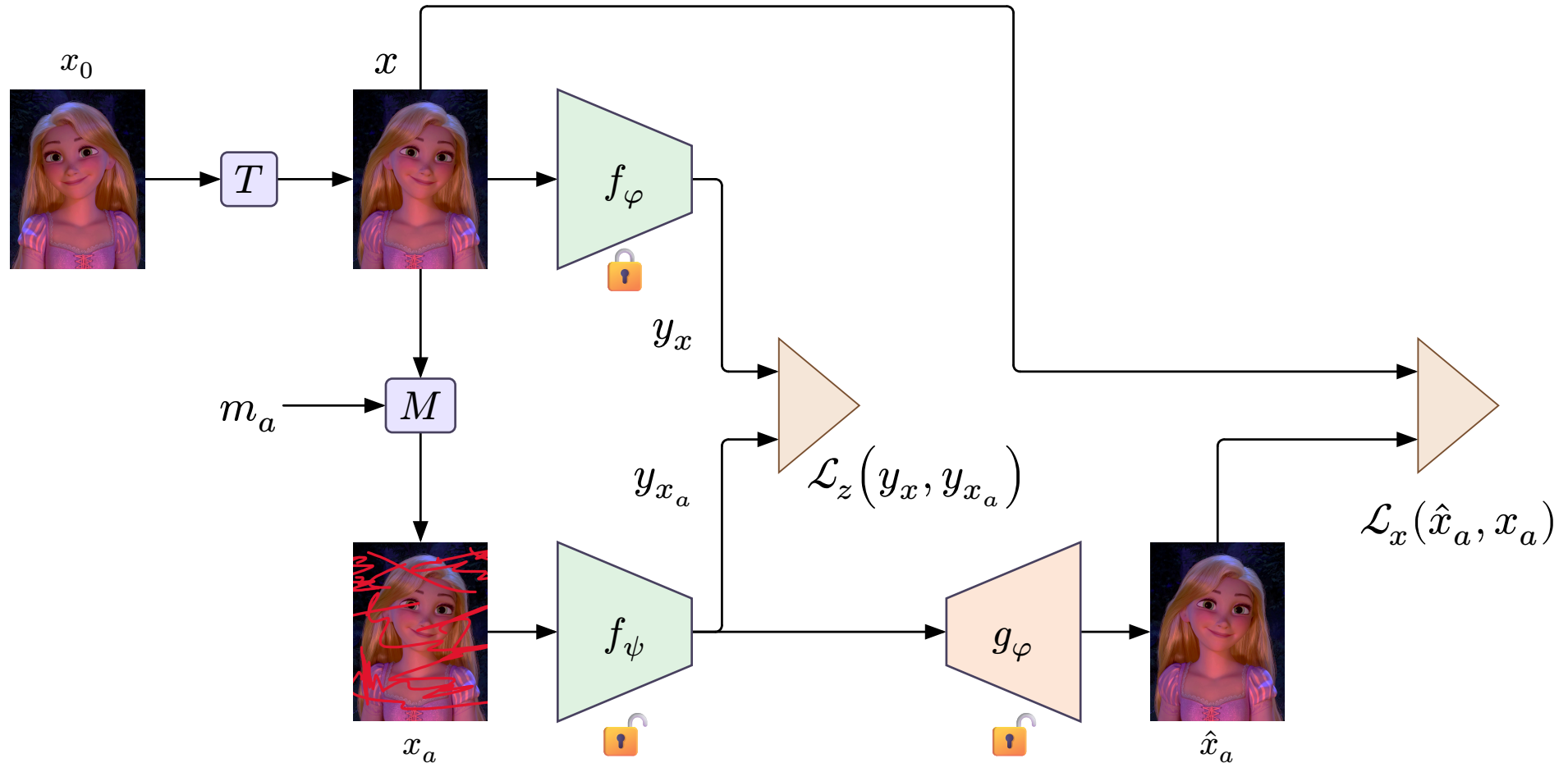
Embedding Reconstruction (AE)

- No mode collapse
- High complexity
- No good loss for pixel space

Embedding Reconstruction (CLIP)

- Mode collapse (many images can have the same embedding)
- Low complexity
- Good loss for latent space

Hybrid approaches





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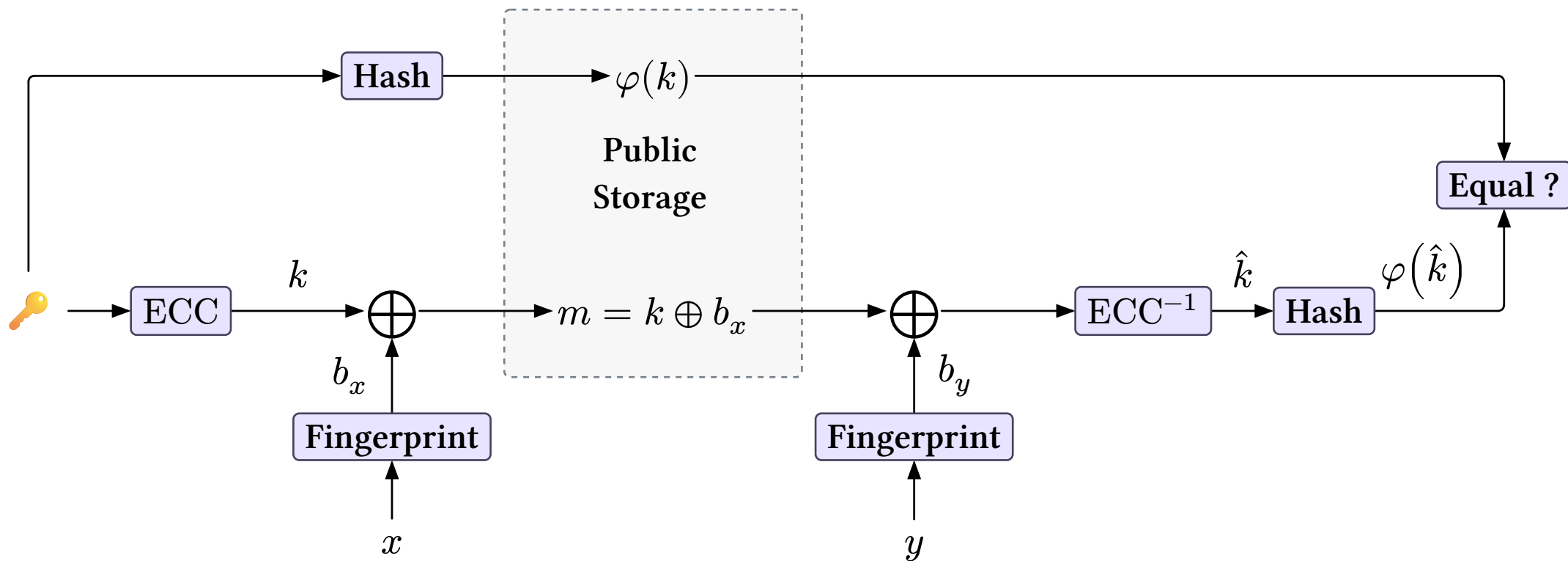
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Traditional privacy protection

- Data owner (protect data) (noise addition, data removal, dim reduction)
- Data user (protect requests) (anonymization, randomized rules)

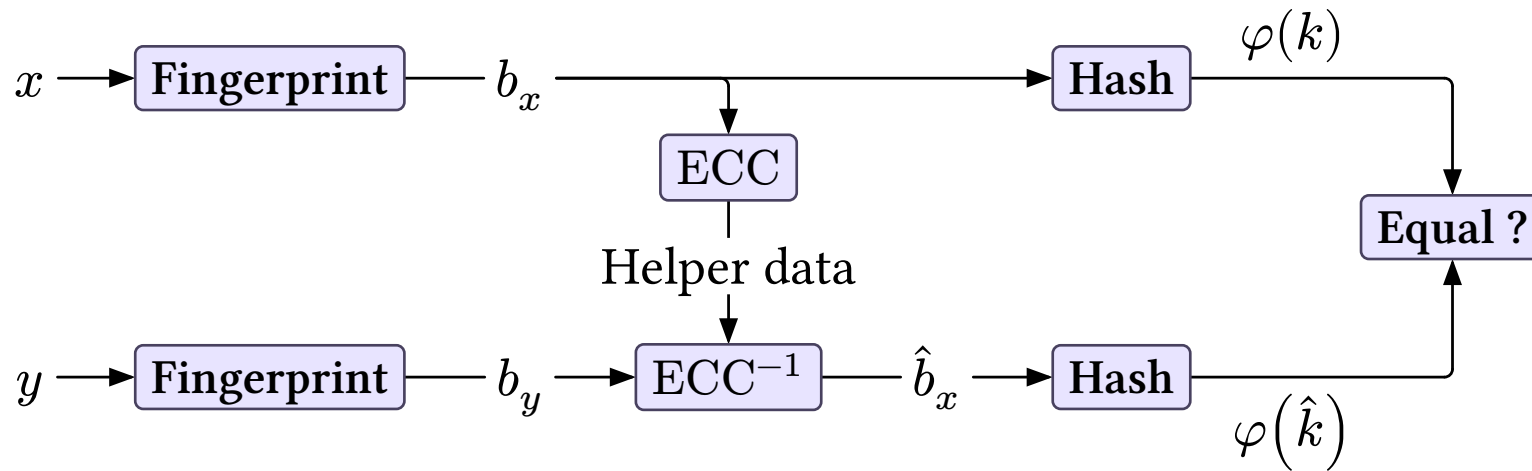


Fuzzy Commitment





Helper data



Redundancy is proportional to $LH_2(P_b)$!